PROJECT POSTER 1 OF 4 · RESEARCH SUMMARY

Lexi AI

Hybrid Retrieval · RAG · Local LLM · FAISS + BM25 · BERT NER · Audit-ready

Business document intelligence: cited answers, compliance checks and clause extraction — running on your own machine.

98.4% Recall@5, hybrid retrieval (63 questions · 5 sectors)

0.92 Mean reciprocal rank (keyword-only: 0.86)

0.84 F1 of the ANN sentence tagger (7 contract tags)

163 automated tests, CI on every push

01 OBJECTIVE

Build a **private, auditable** platform that turns contracts, reports and SOPs into executive briefings, compliance verdicts, extracted clauses and **answers that cite their source** — without sending documents to a third-party cloud — then **measure honestly** how well its retrieval, refusal and classification components work.

02 HYPOTHESIS

H1 — Hybrid search beats keyword-only search
Blending BM25 keyword scores with dense-vector similarity finds the gold passage more often than BM25 alone.
SUPPORTED

H2 — A similarity threshold enables honest refusals
A relevance cut-off lets the system answer "not found" on unanswerable questions instead of guessing.
SUPPORTED · NEEDS TUNING

H3 — A small neural network can tag contract sentences
On 384-d sentence embeddings, a compact ANN, CNN or LSTM labels 7 sentence types far above the 14% chance level.
SUPPORTED · ANN BEST

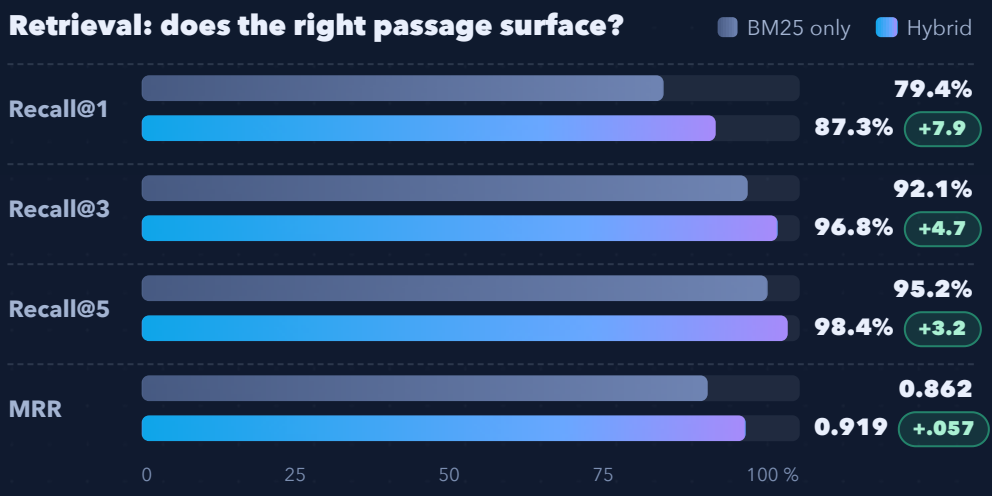
03 PROCEDURE

1. **Ingest & guard** — PDF · DOCX · XLSX · PPTX · CSV · EML · audio → text with **true page provenance**; upload validation and a prompt-injection scan.
2. **Pre-process** — Clean, tokenise, remove stop-words, lemmatise (NLTK[10]).
3. **Index** — 3-sentence overlapping chunks → MiniLM[3,4] 384-d embeddings → **FAISS**[2] vector index + **BM25**[1] keyword index.
4. **Retrieve** — Blend both scores; return the top-5 passages with document, page and relevance.
5. **Reason (RAG)**[5] — A local Llama 3.2 3B[9] answers **only from retrieved passages** with numbered citations; below the relevance threshold it refuses.
6. **Analyse** — Compliance checklist · BERT NER[6,7] + clause rules · ANN / CNN / LSTM[8] tagger · briefings · action items.
7. **Govern** — Role-based access, **hash-chained audit log**, optional zero-retention mode.

RETRIEVAL & REFUSAL TEST
5 sector sets (consulting, finance, healthcare, HR/ops, legal): **63 answerable + 22 unanswerable** questions. Gold = right document + page + text.

TAGGER TEST
280 hand-written sentences, 7 tags × 40. Split: 218 train · 31 val · **31 test**. 80 epochs, Adam.

04 RESULTS



Recall@1 by sector	Consulting	Finance	Health	HR/Ops	Legal
BM25 only	58.3	83.3	81.8	100	77.8
Hybrid	**75.0**	83.3	**90.9**	100	**88.9**



Amber line = chance level (14.3%, 7 classes). The LSTM trails because a sentence embedding is not truly a sequence.

95.4% **Refusal accuracy**, hybrid at its best threshold (answer vs refuse). BM25-only: 54.5%.

75.5% Same metric at the **default** 0.30 threshold — the default needs tuning.

98.3% **Compliance** status accuracy of the offline rule scan vs labelled expectations.

05 CONCLUSION

- **Hybrid retrieval is a clear win** — better Recall@1 than keyword-only in 3 of 5 sectors, never worse on any measure; +7.9 pts overall.
- **Vector similarity makes honest refusals possible**; BM25 scores alone cannot separate answerable from unanswerable questions.
- **A simple ANN is the best tagger** here; the LSTM adds cost without benefit.
- **Limits:** small evaluation sets; thresholds tuned on the same data (optimistic); a 31-sentence test set; and a cautious 3B local LLM — across eight live compliance runs on a 10-page contract it passed 0–4 of 6 checks (about 2 on average) and routed the rest to human review; a verified quote proves a passage exists, not that it proves the point.
- **Next:** blind, larger evaluation · SSO · multi-tenancy · hosted deployment and monitoring.

06 REFERENCES

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